

Question Number	Answer	Additional guidance	Mark
2(a)(i)	The only correct answer is A P <i>B is not correct because the pituitary gland is not involved in the control of the heartbeat</i> <i>C is not correct because the hypothalamus is not involved in the control of the heartbeat</i> <i>D is not correct because the cerebral cortex is not involved in the control of the heartbeat</i>		(1)

Question Number	Answer	Additional guidance	Mark
2(a)(ii)	The only correct answer is D T <i>A is not correct because the medulla is not involved in the control of balance and coordination of movement</i> <i>B is not correct because the pituitary gland is not involved in the control of balance and coordination of movement</i> <i>C is not correct because the hypothalamus is not involved in the control of balance and coordination of movement</i>		(1)

Question Number	Answer	Additional guidance	Mark
2(a)(iii)	The only correct answer is B (Q and R) <i>A is not correct because the medulla is not involved in the negative feedback control of plasma concentration</i> <i>C is not correct because the medulla is not involved in the negative feedback control of plasma concentration</i> <i>D is not correct because the cerebrum is not involved in the negative feedback control of plasma concentration</i>		(1)

Question Number	Answer	Additional guidance	Mark
2(b)(i)	The only correct answer is B dopamine is a neurotransmitter <i>A is not correct because dopamine cannot pass from bloodstream into neurones in the brain</i> <i>C is not correct because high levels of dopamine are not linked to depression, low levels of the chemical are linked to Parkinson's disease</i> <i>D is not correct because dopamine does not cause the breakdown of myelin</i>		(1)

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	An explanation that includes three of the following points: • depression caused by lack of serotonin (1) • SSRI's bind to {channel / proteins} which reuptake serotonin (1) • (therefore) {inhibition of / decreased / prevents} serotonin {(re-)uptake / reabsorption} (1) • serotonin (can still) bind to receptors on post-synaptic membrane (1) • (resulting in more) {depolarisation of / action potentials in} {post-synaptic membrane / relay neurone} (1)	ignore converse ACCEPT SSRIs block protein channels ACCEPT serotonin is not removed from the synapse / serotonin levels remain high in synapse ACCEPT post synaptic receptors ACCEPT opening of Na⁺ voltage gated channels (in post-synaptic membrane) if they refer to post-synaptic in mp4 no need to repeat in mp5	(3)

